

DO NOT WRITE ON TEST



**Financial Math
& Analysis Concepts
~OPEN EVENT~**

190

STATE 2025

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. For any interest calculations, assume a 360-day year.
5. Unless otherwise stated, assume that all plant assets were purchased on January 1.
6. Unless otherwise stated, assume that the overtime rate is 1.5 regular time pay for any hours worked in excess of 40 hours per week.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. A savings account with a balance of \$4,000 earns 3% annual interest compounded annually. What is the balance after 2 years?
 - A. \$4,240.60
 - B. \$4,242
 - C. \$4,243.60
 - D. \$4,250
2. For a loan of \$10,000 with a 7% annual interest rate compounded annually, what is the total amount after 3 years?
 - A. \$10,927.21
 - B. \$11,295.07
 - C. \$11,763.49
 - D. \$12,250.43
3. A loan of \$2,500 is to be repaid with a 4% interest rate annually over 2 years. What is the total amount repaid?
 - A. \$2,604
 - B. \$2,650
 - C. \$2,704
 - D. \$2,800
4. A company forecasts a 10% increase in sales from \$50,000. What are the projected sales?
 - A. \$55,000
 - B. \$60,000
 - C. \$65,000
 - D. \$75,000
5. A customer deposits \$1,000 in a savings account with a 3% annual interest rate compounded yearly. What is the balance after 3 years?
 - A. \$1,092.73
 - B. \$1,093.27
 - C. \$1,100
 - D. \$1,150
6. A retail store operates with a gross profit margin of 40% on sales of \$30,000. Calculate the cost of goods sold.
 - A. \$12,000
 - B. \$18,000
 - C. \$20,000
 - D. \$22,000

7. A loan of \$20,000 is made with a 3% annual interest rate. If the loan is to be repaid in 5 years, what is the total interest paid?
 - A. \$3,000.00
 - B. \$3,185.48
 - C. \$4,000.03
 - D. \$4,500.76
8. A business's inventory value is \$10,000. If it increases by 15% due to new stock arrival, what is the new inventory value?
 - A. \$11,000
 - B. \$11,500
 - C. \$12,000
 - D. \$12,500
9. A savings account offers an interest rate of 2.5% compounded annually. With an initial deposit of \$5,000, what is the balance after 2 years?
 - A. \$5,253.13
 - B. \$5,306.25
 - C. \$5,400
 - D. \$5,500
10. A retailer buys a batch of electronics for \$15,000. If the retailer desires a markup of 20%, what is the selling price?
 - A. \$17,000
 - B. \$17,500
 - C. \$18,000
 - D. \$18,500
11. A bond pays 4% interest annually on a principal of \$2,000. How much interest is earned in 3 years?
 - A. \$240
 - B. \$250
 - C. \$260
 - D. \$280
12. Sarah's credit card statement indicated a balance of \$800, new charges of \$200, a membership fee of \$60, a finance charge of \$10, and a payment of \$500. What is the new balance?
 - A. \$540
 - B. \$550
 - C. \$560
 - D. \$570

13. A previous balance of \$950 was noted on John's credit card, with new charges of \$300, a finance charge of \$20, a late payment fee of \$25, and a payment of \$400. What is the new balance?
- A. \$885
 - B. \$890
 - C. \$895
 - D. \$900
14. Mia's credit card showed a previous balance of \$420.35, new purchases totaling \$189.65, a finance charge of \$6.78, and she made a payment of \$300. What is her new balance?
- A. \$310.35
 - B. \$316.35
 - C. \$316.78
 - D. \$325.78
15. A school purchased a site license for educational software that retails for \$29.99 per license. They paid \$45,000 to use software on their 1,800 laptops. How much did they save by not buying the site license?
- A. \$8,982
 - B. \$9,982
 - C. \$10,982
 - D. \$11,982
16. A corporation acquired a site license for security software that retails for \$75 per installation. They paid \$112,500 for use on 2,000 of their devices. How much did they save by buying the site license?
- A. \$27,500
 - B. \$37,500
 - C. \$62,500
 - D. \$75,000
17. The Smiths calculate their annual home expenses to be: property tax, \$3,200; insurance, \$550; utilities, \$1,700; maintenance, \$1,300; mortgage interest, \$8,000; lost interest on down payment, \$600. They anticipate an income tax saving of \$2,200. What is the net cost of home ownership for the first year?
- A. \$12,150
 - B. \$13,150
 - C. \$13,250
 - D. \$14,150

18. For the Williams family, the first year home operating expenses are projected as: property tax, \$4,000; insurance, \$500; utilities, \$2,000; maintenance, \$1,500; mortgage interest, \$9,000; lost interest on down payment, \$700. Their expected income tax savings amount to \$2,500. What is the net cost of owning their home in the first year?
 - A. \$13,200
 - B. \$14,200
 - C. \$15,200
 - D. \$16,200

19. Mike had \$200 in change in his cash register at the start of the day. At the end of his shift, the cash register showed \$3,240.50. The register totals indicated that he received \$3,265.75 and paid out \$220.25 during the day. How much was the cash over or short?
 - A. \$5.00 over
 - B. \$5.00 short
 - C. \$25.25 over
 - D. \$25.25 short

20. Jessica started with \$165.25 in her cash register. At the end of her workday, there was \$4,115.45 in the register. According to the register totals, she received \$4,056.20 and paid out \$100.00 during the shift. How much was the cash over or short?
 - A. \$6.00 over
 - B. \$6.00 short
 - C. \$59.25 over
 - D. \$59.25 short

21. Emma earned \$4,500 last month. She paid an overall FICA tax rate of 7.65% that consists of a 6.2% social security tax and a 1.45% Medicare tax. Federal income taxes withheld from her wages were \$920. What total amount of taxes was deducted from her pay last month?
 - A. \$344.25
 - B. \$1,084.25
 - C. \$1,264.25
 - D. \$2,420.25

22. Alex earned \$5,000 last month. He paid an overall FICA tax rate of 7.65% that consists of a 6.2% social security tax and a 1.45% Medicare tax. Federal income taxes withheld from his wages were \$1,100. What total amount of taxes was deducted from his pay last month?
 - A. \$382.50
 - B. \$1,100.00
 - C. \$1,482.50
 - D. \$2,482.50

23. A local business pays \$1,200 for a front-page banner in a community newsletter delivered to 4,800 homes. What is the cost of the ad per household?
- A. \$0.20
 - B. \$0.25
 - C. \$0.30
 - D. \$0.50
24. An online retailer sponsors a catalog insert in a regional publication with a circulation of 15,000 homes, at a cost of \$3,000 per publication. What is the cost per household?
- A. \$0.10
 - B. \$0.15
 - C. \$0.20
 - D. \$0.25
25. A restaurant advertises its grand opening in a local paper, spending \$750 for an insert that reaches 3,000 households. What is the ad cost per household?
- A. \$0.20
 - B. \$0.25
 - C. \$0.30
 - D. \$0.50
26. Two developers created a 5,000-line application. The first developer worked for 300 hours at \$25 per hour, while the second spent 250 hours and was paid \$20 an hour. What was the average development cost per line of the application?
- A. \$2.50
 - B. \$2.75
 - C. \$4.00
 - D. \$5.50
27. A team of two engineers wrote an 8,000-line software program. One engineer worked 420 hours at \$22 per hour, and the other worked 400 hours at \$18 per hour. What was the average cost per line for this software program?
- A. \$2.06
 - B. \$2.55
 - C. \$3.85
 - D. \$4.20
28. For a 7,200-line system, one programmer worked 350 hours at \$19.50 per hour, and another worked 340 hours at \$23.50 per hour. What was the average cost per line to develop this system?
- A. \$2.06
 - B. \$3.65
 - C. \$4.50
 - D. \$5.25

29. Alex plans to rent a studio for \$850 a month, which includes utilities. The security deposit is one month's rent. He estimates additional annual costs: insurance \$150; gym access, \$300; parking space, \$180. What is Alex's first-year cost of renting the studio?
- A. \$11,280
 - B. \$11,430
 - C. \$11,680
 - D. \$11,730
30. Bella intends to rent a condo for \$950 per month, including water. The security deposit will be one month's rent. Her other annual costs include insurance \$200; cable, \$720; parking, \$300. What is Bella's first-year cost of renting the condo?
- A. \$12,920
 - B. \$13,170
 - C. \$13,570
 - D. \$14,070
31. Nina had 45 square yards of ceramic tile installed in her bathroom. The tile cost \$9.50 per square yard, the waterproofing membrane cost \$5.75 per square yard, and the installation cost \$7 per square yard. The sales tax rate is 7%, applied only to materials. What was the total cost of her project?
- A. \$990.75
 - B. \$1,049.29
 - C. \$1,235.25
 - D. \$1,357.34
32. The monthly premium for vision insurance provided by an employer is \$24 per employee. If employees are responsible for 50% of the premium, what is the total annual amount paid by an employee?
- A. \$194.00
 - B. \$288.00
 - C. \$432.00
 - D. \$576.00
33. James sold 150 shares of a mutual fund at \$15.75 per share. His initial investment in the shares was \$2,250. What was his profit or loss?
- A. \$112.50 profit
 - B. \$112.50 loss
 - C. \$237.50 profit
 - D. \$237.50 loss

34. Mia and Ethan's combined gross income for the year was \$120,000. They took a standard deduction of \$12,000 and 3 exemptions at \$4,500 each to determine their taxable income. What was it?
- A. \$95,500
 - B. \$94,500
 - C. \$106,500
 - D. \$110,000
35. A jeweler bought watches to sell for \$200 and sold those watches for \$280. If the markup is based on cost, what markup percentage is the jeweler charging?
- A. 25%
 - B. 30%
 - C. 40%
 - D. 50%
36. Emily listed these items on her checking account deposit slip: (bills) 25 ones, 4 fives, 3 tens, and 5 twenties; (coins) 156 quarters, 85 dimes, 33 nickels; one check for \$60.50. What was the total of Emily's deposit?
- A. \$300.25
 - B. \$284.65
 - C. \$400.25
 - D. \$450.75
37. Mike listed these items on his checking account deposit slip: (bills) 60 ones, 6 fives, 5 tens, and 7 twenties; (coins) 200 quarters, 100 dimes, 40 nickels; one check for \$72.50. What was the total of Mike's deposit?
- A. \$400.00
 - B. \$450.50
 - C. \$513.50
 - D. \$414.50
38. What are the gross earnings for an employee who works 45 hours and is paid \$10.20/hour?
- A. \$459.00
 - B. \$499.50
 - C. \$510.00
 - D. \$561.00
39. What are the gross earnings for an employee who works 50 hours and is paid \$13.45/hour?
- A. \$647.50
 - B. \$672.50
 - C. \$707.50
 - D. \$767.50

40. What are the gross earnings for an employee who works 30 hours and is paid \$12.00/hour?
- A. \$320.00
 - B. \$360.00
 - C. \$396.00
 - D. \$420.00
41. Sara earns a regular wage of \$13.25 per hour. Overtime is paid at time-and-a-half, and work on Saturdays is double time. Her hours last week were: Monday 8 hours, Tuesday 8 hours, Wednesday 8 hours, Thursday 9 hours, Friday 9 hours, Saturday 3 hours. What was her gross pay?
- A. \$560.25
 - B. \$612.75
 - C. \$649.25
 - D. \$732.50
42. Owen's regular hourly rate is \$20. He earns time-and-a-half for any hours beyond 40 and double the time for work on Sundays. Last week, he worked: Monday 8 hours, Tuesday 8 hours, Wednesday 8 hours, Thursday 8 hours, Friday 7 hours, Sunday 5 hours. What was his gross pay for the week?
- A. \$840.00
 - B. \$930.00
 - C. \$980.00
 - D. \$1,100.00
43. A pound of organic coffee beans is priced at \$15.99. A half pound of the same coffee is \$8.50. How much can be saved by buying 1 pound of coffee instead of buying two half pounds?
- A. \$0.81
 - B. \$0.91
 - C. \$1.01
 - D. \$1.21
44. A 2-liter bottle of soda is priced at \$1.75. A 1-liter bottle of the same soda is \$1.20. How much can be saved by buying 1 2-liter bottle instead of two 1-liter bottles?
- A. \$0.65
 - B. \$0.75
 - C. \$0.85
 - D. \$0.95

45. A 5-pound bag of flour is priced at \$4.50. A 2.5-pound bag of the same flour is \$2.75. How much can be saved by buying 1 5-pound bag instead of two 2.5-pound bags?
- A. \$1.00
 - B. \$1.10
 - C. \$1.20
 - D. \$1.30
46. A 16-ounce jar of peanut butter is priced at \$3.99. An 8-ounce jar of the same peanut butter is \$2.50. How much can be saved by buying 1 16-ounce jar instead of two 8-ounce jars?
- A. \$0.91
 - B. \$1.01
 - C. \$1.21
 - D. \$1.31
47. You took out a car loan for \$10,000 for 2 years at an annual interest rate of 7.5%. What is the total amount of interest owed on the loan?
- A. \$750
 - B. \$1,000
 - C. \$1,500
 - D. \$2,000
48. You borrowed \$2,500 for 18 months at an annual interest rate of 9%. What is the total amount of interest owed on the loan?
- A. \$168.75
 - B. \$225.50
 - C. \$337.50
 - D. \$450.75
49. Last quarter, Germany had exports of \$320.5 billion and imports of \$299.8 billion. How much of a deficit or surplus did they have?
- A. \$20.7 billion deficit
 - B. \$20.7 billion surplus
 - C. \$21.7 billion deficit
 - D. \$21.7 billion surplus
50. Crafty Tables Co. plans to make designer desks and sell them at \$200 each. They estimate their fixed costs to produce the desks at \$100,000 and variable costs of \$125 per desk. How many desks must Crafty Tables Co. sell to break even?
- A. 500
 - B. 800
 - C. 1250
 - D. 1334